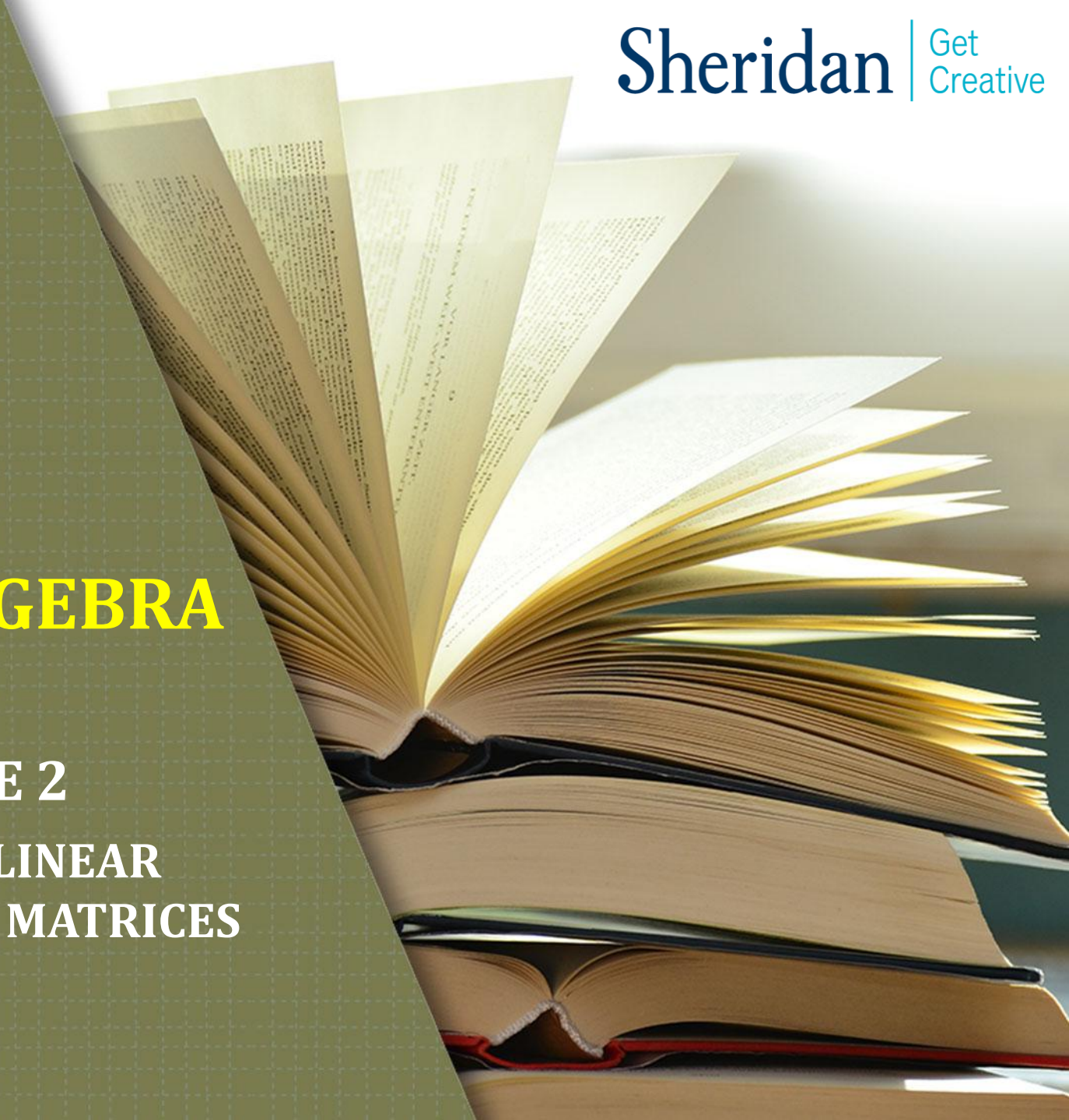


LINEAR ALGEBRA

LECTURE 2

SYSTEMS OF LINEAR EQUATIONS AND MATRICES



Introduction to Systems of Linear Equations



Generally, we define a linear equation in the n variables,

$$x_1, x_2, \dots, x_n$$

in the form of,

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

Where $a_1, a_2, \dots, a_n$ and b are constants, and the a 's are not all zero.

In the special case where $b = 0$, a linear equation in the n variables $x_1, x_2, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = 0$$

This is called a **homogeneous linear equation**.

A general linear system of m equations in the n unknowns $x_1, x_2, \dots, x_n$ can be written,

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

A solution of a linear system in n unknowns $x_1, x_2, \dots, x_n$ is a sequence of n numbers $s_1, s_2, \dots, s_n$

$$x_1 = s_1$$

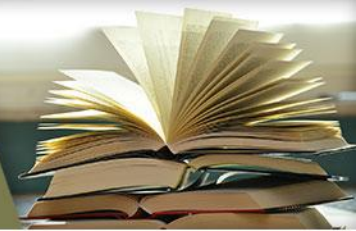
$$x_2 = s_2$$

$$\vdots$$

$$x_n = s_n$$

this solution can be written as $(s_1, s_2, \dots, s_n)$ which is called an **n-tuple** (ordered list of n items).

Linear Systems in Two and Three Unknowns



Consider the linear system,
$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

there are three possible solutions.

1. The lines may be parallel and distinct
2. The lines may intersect at only one point
3. The lines may coincide



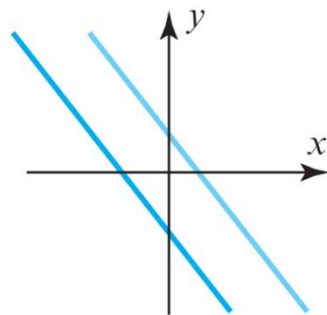
no solution



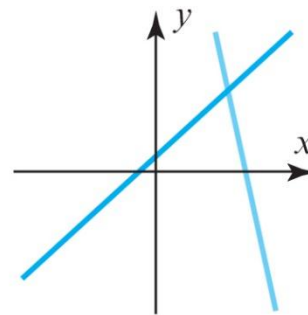
one solution



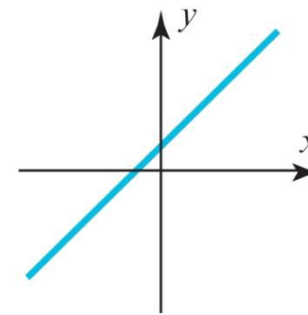
many solutions



No solution



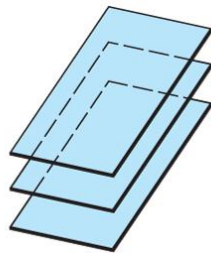
One solution



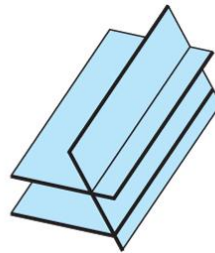
Infinitely many
solutions
(coincident lines)

Note: In general, a linear system is **consistent** if it has at least one solution and **inconsistent** if it has no solutions.

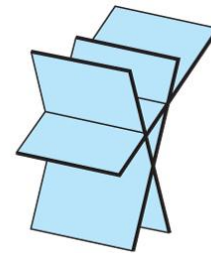
Now, Could you speculate the solution of a linear system of three equations? (Discuss about the possible solutions)



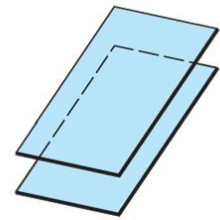
No solutions
(three parallel planes;
no common intersection)



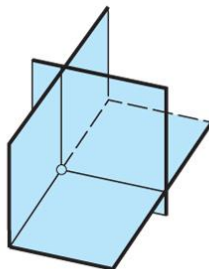
No solutions
(two parallel planes;
no common intersection)



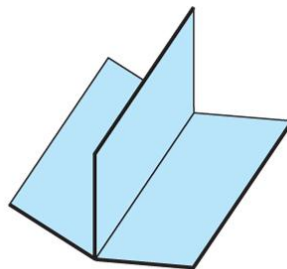
No solutions
(no common intersection)



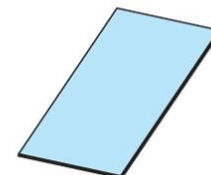
No solutions
(two coincident planes
parallel to the third;
no common intersection)



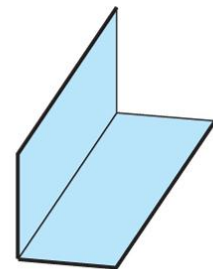
One solution
(intersection is a point)



Infinitely many solutions
(intersection is a line)



Infinitely many solutions
(planes are all coincident;
intersection is a plane)



Infinitely many solutions
(two coincident planes;
intersection is a line)



➤ Exercise 1

Which following equations are not linear?

$$\sqrt{x_1} + 2x_2 + x_3 = 1$$

$$x + 3y = 7$$

$$\frac{1}{2}x + 3z = -1$$

$$\sin x + y = 0$$

$$x + 3y^2 = 4$$

➤ Exercise 2

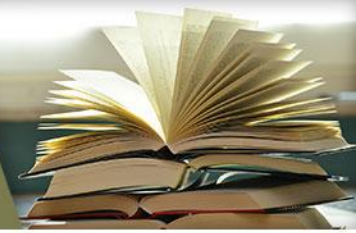
Solve the following linear systems.

$$1. \begin{cases} x + y = 4 \\ 3x + 3y = 6 \end{cases}$$

$$2. \begin{cases} 4x - 2y = 1 \\ 16x - 8y = 4 \end{cases}$$

$$3. \begin{cases} x - y + 2z = 5 \\ 2x - 2y + 4z = 10 \\ 3x - 3y + 6z = 15 \end{cases}$$

Augmented Matrices and Elementary Row Operations



Now, by mentally keeping track of the location of the +’s, the x’s, and the =’s in the linear system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

we can abbreviate the system by writing the rectangular array of numbers

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & & a_{2n} & b_2 \\ & \vdots & \ddots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix} \Rightarrow \text{augmented matrix}$$



The basic method for solving a linear system is to perform some algebraic operations on the system that do not alter the solution set. Typically, the algebraic operations are:

1. Multiply an equation through by a nonzero constant.
2. Interchange two equations.
3. Add a constant times one equation to another.

As you know, the rows of an augmented matrix depict the equations in the associated system,

1. Multiply a row through by a nonzero constant.
2. Interchange two rows.
3. Add a constant times one row to another.



elementary row operations

To clarify the tips above, we will bring an example.


$$\begin{cases} x + y + 2z = 9 \\ 2x + 4y - 3z = 1 \\ 3x + 6y - 5z = 0 \end{cases} \quad \text{so} \quad \begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

Add -2 times the first equation to the second to obtain,

$$\begin{cases} x + y + 2z = 9 \\ 2y - 7z = -17 \\ 3x + 6y - 5z = 0 \end{cases} \quad \text{so} \quad \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

Add -3 times the first equation to the third to obtain,

$$\begin{cases} x + y + 2z = 9 \\ 2y - 7z = -17 \\ 3y - 11z = -27 \end{cases} \quad \text{so} \quad \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

Multiply the second equation by $1/2$ to obtain,


$$\begin{cases} x + y + 2z = 9 \\ y - \frac{7}{2}z = -\frac{17}{2} \\ 3y - 11z = -27 \end{cases} \quad \text{so} \quad \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

Add -3 times the second equation to the third to obtain,

$$\begin{cases} x + y + 2z = 9 \\ y - \frac{7}{2}z = -\frac{17}{2} \\ -\frac{1}{2}z = -\frac{3}{2} \end{cases} \quad \text{so} \quad \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{bmatrix}$$

Multiply the third equation by -2 to obtain,

$$\begin{cases} x + y + 2z = 9 \\ y - \frac{7}{2}z = -\frac{17}{2} \\ z = 3 \end{cases} \quad \text{so} \quad \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Add -1 times the second equation to the first to obtain,


$$\begin{cases} x + \frac{11}{2}z = \frac{35}{2} \\ y - \frac{7}{2}z = -\frac{17}{2} \\ z = 3 \end{cases} \quad \text{so} \quad \begin{bmatrix} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Add $-11/2$ times the third equation to the first and $7/2$ times the third equation to the second to obtain,

$$\begin{cases} x = 1 \\ y = 2 \\ z = 3 \end{cases} \quad \text{so} \quad \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

➤ Exercise 3

Solve this system by using elementary row operations.

$$\begin{cases} 2x - 4y - z = 1 \\ x - 3y + z = 1 \\ 3x - 5y - 3z = 1 \end{cases}$$

Gaussian Elimination



In the former example, we solved a linear system in the unknowns x , y , and z by reducing the augmented matrix to the form,

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

To be of this form, a matrix must have the following properties:

1. If a row does not consist entirely of zeros, then the first nonzero number in the row is a 1. We call this a leading 1.
2. If there are any rows that consist entirely of zeros, then they are grouped together at the bottom of the matrix.
3. In any two successive rows that do not consist entirely of zeros, the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row.

4. Each column that contains a leading 1 has zeros everywhere else in that column.

A matrix that has the first three properties is said to be in **row echelon form (REF)**.

The following matrices are in row echelon form (it looks like a staircase):

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}$$

The following matrices are in **reduced row echelon form (RREF)** (it looks like a staircase and the pivot columns, the columns that contain a leading 1, are cleaned up).

$$\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

As shown in the examples above, a matrix in **row echelon** form has **zeros below each leading 1**, whereas a matrix in **reduced row echelon** form has zeros **below and above each leading 1**. Thus, with any real numbers substituted for the *'s, all matrices of the following types are in row echelon form:

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

All matrices of the following types are in reduced row echelon form:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$



Note If, by a sequence of elementary row operations, the augmented matrix for a system of linear equations is put in reduced row echelon form, then, the solution set can be obtained either by inspection or by converting certain linear equations to parametric form.

To illustrate the note above, let's consider the following matrix.

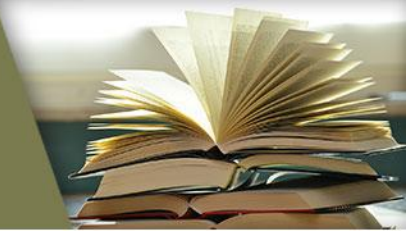
$$\begin{bmatrix} 1 & -5 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We can find,

$$x - 5y + z = 4 \quad \text{so} \quad x = 5y - z + 4$$

From this equation we see that the free variables can be assigned arbitrary values, say $y = s$ and $z = t$, which can determine the value of x .

Elimination Methods



We will start by considering this matrix,

$$\begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

Step 1. Locate the leftmost column that does not consist entirely of zeros.

$$\begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

Step 2. Interchange the top row with another row, if necessary, to bring a nonzero entry to the top of the column found in Step 1.



$$\begin{bmatrix} 2 & 4 & -10 & 6 & 12 & 28 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

The first and second rows were interchanged.

Step 3. If the entry that is now at the top of the column found in Step 1 is a , multiply the first row by $1/a$.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

The first row was multiplied by $1/2$.

Step 4. Add suitable multiples of the top row to the rows below so that all entries below the leading 1 become 0.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{bmatrix}$$

-2 times the first row was added to the third row.

Step 5. Now cover the top row in the matrix and begin again with Step 1 applied to the sub-matrix. You must continue until the entire matrix is in row echelon form.


$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{bmatrix}$$

Leftmost nonzero column in the sub-matrix.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -7/2 & -6 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{bmatrix}$$

First row in the sub-matrix was multiplied by $-1/2$.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -7/2 & -6 \\ 0 & 0 & 0 & 0 & 1/2 & 1 \end{bmatrix}$$

-5 times the first row was added to the second row.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -7/2 & -6 \\ 0 & 0 & 0 & 0 & 1/2 & 1 \end{bmatrix}$$

Leftmost nonzero column in the sub-matrix.


$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -7/2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

First row in the new sub-matrix was multiplied by 2.

Step 6. Beginning with the last nonzero row and working upward, add suitable multiples of each row to the rows above to introduce zeros above the leading 1's.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

7/2 times the third row was added to the second row.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

-6 times the third row was added to the first row.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

5 times the second row was added to the first row.


$$\begin{bmatrix} 1 & 2 & -5 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

Pivot variables: x_1, x_3, x_5

Free variables: x_2, x_4

If assume $x_2 = s$ and $x_4 = t$, then

$$x_1 = 7 - 2s - 3t$$

$$x_2 = s$$

$$x_3 = 1$$

$$x_4 = t$$

$$x_5 = 2$$

The algorithm we have just described for reducing a matrix to reduced row echelon form is called **Gauss-Jordan elimination**. It consists of two parts, a **forward** phase in which zeros are introduced **below** the leading 1's and a **backward** phase in which zeros are introduced **above** the leading 1's.



➤ Exercise 4

Solve by Gauss–Jordan elimination.

$$x_1 + 3x_2 - 2x_3 + 2x_5 = 0$$

$$2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1$$

$$5x_3 + 10x_4 + 15x_6 = 5$$

$$2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6$$

➤ Exercise 5

Use Gauss–Jordan elimination to solve the homogeneous linear system.

$$x_1 + 3x_2 - 2x_3 + 2x_5 = 0$$

$$2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = 0$$

$$5x_3 + 10x_4 + 15x_6 = 0$$

$$2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 0$$



THEOREM 1

If a homogeneous linear system has n unknowns, and if the reduced row echelon form of its augmented matrix has r nonzero rows, the system has $n - r$ free variables.

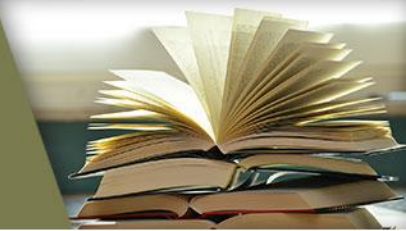
THEOREM 2

Homogeneous linear system with more unknowns than equations has infinitely many solutions

There are three facts about row echelon forms:

1. Every matrix has a unique reduced row echelon form.
2. Row echelon forms are not unique.
3. The reduced row echelon and all row echelon forms of a matrix A have the same number of zero rows, and the leading 1's always occur in the same positions. Those are called the pivot positions of A .

Matrices and Matrix Operations



A matrix is a rectangular array of numbers and the numbers in the array are called the entries in the matrix.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \\ -1 & 4 \end{bmatrix}, \quad B = [2 \quad 1 \quad -3]$$

Size of a matrix

The size of a matrix is described in terms of the number of rows and columns it contains.

Row or column matrix

A matrix with only one row, is called a row vector and a matrix with only one column, is called a column vector.



A $m \times n$ matrix

Thus a $m \times n$ matrix might be written as

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

Equality of matrices

Two matrices are defined to be equal if they have the same size and their corresponding entries are equal.

Addition and subtraction

Consider the matrices,

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 2 \\ 4 & -2 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} -4 & 3 & 5 \\ 2 & 2 & 0 \\ 3 & 2 & -4 \end{bmatrix}$$

$$A + B = \begin{bmatrix} -2 & 4 & 5 \\ 1 & 2 & 2 \\ 7 & 0 & 3 \end{bmatrix} \quad A - B = \begin{bmatrix} 6 & -2 & -5 \\ -3 & -2 & 2 \\ 1 & -4 & 11 \end{bmatrix}$$

Scalar multiples

If A is any matrix and c is any scalar, then the product cA is the matrix obtained by multiplying each entry of the matrix A by c .

Multiplying matrices

If A is an $m \times r$ matrix and B is an $r \times n$ matrix, then:

To find the entry in row i and column j of $A \times B$, single out row i from the matrix A and column j from the matrix B . Multiply the corresponding entries from row and column together, and then add up the resulting products. Consider the following multiplying,

$$A \times B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \times \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} & & & 13 \\ & & 26 & \end{bmatrix}$$

$$(2 \times 4) + (6 \times 3) + (0 \times 5) = 26$$

$$(1 \times 3) + (2 \times 1) + (4 \times 2) = 13$$

Note The definition of matrix multiplication requires that the number of columns of the first factor A be the same as the number of rows of the second factor B .

THEOREM 3

If A is an $m \times n$ matrix, and if X is an $n \times 1$ column vector, then AX can be expressed as a linear combination of the column vectors of A and the coefficients are the entries of X .

$$AX = \begin{bmatrix} a_{11}x_1 & \cdots & a_{1n}x_n \\ \vdots & \ddots & \vdots \\ a_{m1}x_1 & \cdots & a_{mn}x_n \end{bmatrix} = x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

➤ Exercise 6

Calculate the below.

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \times \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

$$4 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix}$$

Now calculate the following: $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$

Key idea: take one column of B at a time. Its three entries become the coefficients multiplying the three columns of A.

The three columns of A are:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

1st column of B: [4, 0, 2]

$$4 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix}$$

2nd column of B: [1, -1, 7]

$$1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + -1 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 7 \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 27 \\ -4 \end{bmatrix}$$

3rd column of B: [4, 3, 5]

$$4 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 5 \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 30 \\ 26 \end{bmatrix}$$

4th column of B: [3, 1, 2]

$$3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 13 \\ 12 \end{bmatrix}$$

Final answer

$$AB = \begin{bmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{bmatrix}$$



Transpose

If A is any $m \times n$ matrix, then the transpose of A , denoted by A^T , is defined to be the $n \times m$ matrix that results by interchanging the rows and columns of A ; that is, the first column of A^T is the first row of A , the second column of A^T is the second row of A , and so forth.

$$(A^T)_{ij} = (A)_{ji}$$

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 2 \\ 2 & 6 \\ 4 & 0 \end{bmatrix}$$

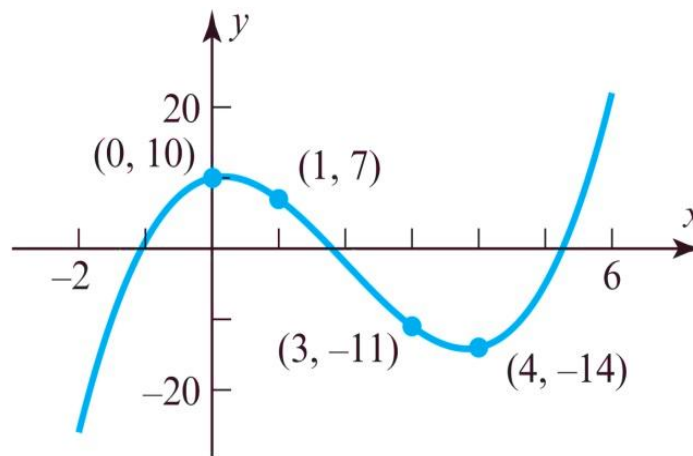
Trace of a matrix

If A is a square matrix, then the trace of A , $tr(A)$, is defined to be the sum of the entries on the main diagonal of A . The trace of A is undefined if A is not a square matrix.

➤ Project: Systems of Linear Equations and Matrices

Find the coefficients a , b , c , and d so that the curve shown in the accompanying figure is the graph of the equation,

$$y = ax^3 + bx^2 + cx + d.$$



Answer: $a = 1$, $b = -6$, $c = 2$

GOOD LUCK

